MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY

COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY

DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION ENGINEERING

Explainable AI Model for Pulmonary Edema Detection in Chest X-Ray Images.

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Explainable AI Model for Pulmonary Edema Detection in Chest X-Ray Images.

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CERTIFICATION

The undersigned, hereby certify that the project titled "Explainable AI Model for Pulmonary Edema Detection in Chest X-rays images," submitted in partial fulfillment of the requirements for the Bachelor of Engineering in Telecommunication Systems at Mbeya University of Science and Technology, has been reviewed and is recommended for acceptance.

Name of the Supervisor: Dr. Charles Nyatega

Signature of the Supervisor:

24–03–2026

DECLARATION

I, "Boniface Ben Sankwa", declare that I carried out the work reported in this report for the Department of Electronics and Telecommunication Engineering at MUST, under the supervision of " Dr. Charles Nyatega ", I solemnly declare that, to the best of my knowledge, no part of this report has been submitted here or elsewhere in a previous application for award or a degree.

All sources of knowledge used have been duly acknowledged.

Name of the Student: Boniface Ben Sankwa

Signature of the Student:

24 -03-2026

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List of Abbreviations

AI	Artificial Intelligence
CNN	Convolutional Neural Network
GAN	Generative Adversarial Network
IEEE	Institute of Electrical and Electronics Engineers
IoT	Internet of Things
NLP	Natural Language Processing
OC	Optical Cup
OCR	Optical Character Recognition
STT	Speech-To-Text
TTS	Text-To-Speech
VR	Virtual Reality
XAI	Explainable Artificial Intelligent
Grad-CAM	Gradient-weighted Class Activation Mapping

Chapter 1: Introduction

1.1 Background Information

Pulmonary edema is a serious medical condition in which fluid accumulates in the lungs, impairing breathing and reducing oxygen exchange. It is often associated with heart failure and can be life-threatening if not detected early(Abbasi et al., 2025; Danilov et al., 2024). Chest X-ray imaging is one of the most widely used diagnostic tools because it is affordable, accessible, and non-invasive (Irvin et al., 2019; Wang et al., 2017).

However, interpretation of chest X-rays is not straightforward and requires experienced radiologists, since pulmonary edema often shares visual similarities with pneumonia and other lung diseases (Rajpurkar et al., 2017). This can lead to diagnostic errors and delays in treatment.

Recent advances in Artificial Intelligence (AI), particularly deep learning, have enabled computers to automatically analyze medical images. Convolutional Neural Networks (CNNs) have demonstrated strong performance in detecting lung diseases from chest X-rays (Esteva et al., 2019; Litjens et al., 2017).

Despite this success, most models act as "black-boxes," providing predictions without explaining the reasoning behind them. This lack of transparency reduces clinical trust. Therefore, there is a need for **Explainable AI (XAI)** systems that can detect pulmonary edema while also highlighting the regions of the chest X-ray used for prediction (Danilov et al., 2023).

1.2 Problem Statement

Despite the widespread use of chest X-rays, the detection of pulmonary edema remains a challenge. Diagnosis relies heavily on skilled radiologists, yet many healthcare facilities face a shortage of trained experts, leading to delays and potential errors (WHO et al., 2022; Topol, 2019).

Pulmonary edema can look a lot like pneumonia. This makes it really tough to figure out what is going on by looking at it. Pulmonary edema is tricky to diagnose because it can be mistaken, for pneumonia, which means there is a chance of making mistakes. (Rajpurkar et al., 2017). Deep learning models are really good. The problem is that we do not understand how they work. This is an issue because it stops doctors from using them to help people. Deep learning models need to be more transparent so that doctors can trust them and use them in their work with learning models. (Danilov et al., 2024). Thus, there is a need for an automated system that not only detects pulmonary edema but also provides visual explanations to support medical decision-making (Abbasi et al., 2025).

1.3 Project objective

The objectives of this project are divided into two main categories, the main objective and the specific objectives.

1.3.1 Main objective

To develop an Explainable AI system for detecting pulmonary edema in chest X-ray images, capable of providing diagnostic results along with visual explanations of the regions used for prediction.

1.3.2 Specific objectives

i. To study the radiographic features of pulmonary edema in chest X-ray images.
ii. To collect and preprocess chest X-ray datasets for training and validation.
iii. To design and train a CNN model for pulmonary edema detection.
iv. To apply Explainable AI techniques (e.g., Grad-CAM) to visualize the regions of the X-ray used for prediction.
v. To evaluate the performance of the system.
vi. To develop a simple interface that displays both predictions and heatmaps for medical practitioners.

1.4 Significant of the Project

Once realized, the proposed project will provide the following advantages

i. Enhances trust in AI systems by providing transparent predictions.
ii. Supports clinical decision-making by offering explanations alongside results.
iii. Reduces the workload of radiologists and medical staff.
iv. Promotes early diagnosis and timely treatment of patients.
v. Provides a solution suitable for hospitals with limited resource.

1.5 Scope of the Project

This project focuses exclusively on chest X-ray images for pulmonary edema detection. It employs CNN-based deep learning techniques combined with Explainable AI methods such as Grad-CAM. Publicly available datasets are used for training and testing (Irvin et al., 2019). The system is designed to support radiologists and not replace clinical judgment.

1.6 Project Limitation

This proposed project has the following limitations

i. The accuracy of the system depends on the quality and size of the dataset (Shorten & Khoshgoftaar, 2019).
ii. Limited computing resources may restrict the use of very deep models (Goodfellow et al., 2016).
iii. Explainability techniques may produce incomplete or imprecise visualizations, requiring clinical validation.
iv. The model is restricted to chest X-rays and may not generalize to other imaging modalities such as CT scans.

1.7 Project Structure

This section of the project report discusses the organization and general structure. The report is divided into six chapters. Also, the chapters are split into multiple subheadings to provide specific information on chapter sections. Chapter 1 is the introductory part of the project. It provides an introduction to the study and also discusses the project's background. This also outlines the project's objectives, significance, scope, limitations, and structure. Chapter 2 contains the literature review. It includes a review of several academic publications on the subject. Documents like journals, journals, and publications have all been reviewed.

Chapter 2: Literature Review

2.1 Introduction

This chapter is about looking at what people have found out about detecting pulmonary edema in chest X-ray pictures how deep learning is used in medical imaging and what Explainable Artificial Intelligence is all about. The point of looking at what people have done is to get a good understanding of the basics see what is good and what is not so good about the old ways and find out what still needs to be figured out which is what this project is trying to do.

Pulmonary edema is a serious condition that needs to be diagnosed quickly and correctly. Chest X-rays are used a lot because they are cheap and easy to get. They can be hard to understand because pulmonary edema can look like other lung diseases, such as pneumonia (Irvin et al., 2019; Rajpurkar et al., 2017). Deep learning something called Convolutional Neural Networks is really good at looking at medical pictures and finding out what is wrong (Abbasi et al., 2025; Litjens et al., 2017).

Even though deep learning is really good it is like a black box that you put something in and get an answer out. You do not know how it got that answer. This is a problem because doctors do not trust it if they do not understand how it works, (Abbasi et al., 2025; Litjens et al., 2017). So some new techniques, like Grad-CAM and saliency maps have been made to help explain what the computer is thinking when it looks at chest X-rays.

So this chapter looks at what people have learned what they have tried and what they have made so that we can make a system that uses Explainable Artificial Intelligence to detect pulmonary edema. The chapter ends by saying what we still need to figure out and why this project is important. Pulmonary edema detection is what we are focusing on and Explainable Artificial Intelligence is what we are using to make it better.

2.2 Conceptual/Theoretical Framework

This section explains the core technologies and theories that support the system proposed in this study. The framework is built around the use of **Convolutional Neural Networks (CNNs)** for image classification and **Explainable Artificial Intelligence (XAI)** for model transparency. These technologies are applied to detect pulmonary edema in chest X-ray images.

The following is the core Technologies Explained;

i. Convolutional Neural Networks (CNNs).

   CNNs are deep learning models designed for image analysis. They automatically extract features from medical images and classify them based on learned patterns. Architectures such as ResNet, DenseNet, and VGG are commonly used due to their high accuracy in medical imaging tasks (Abbasi et al., 2025; Rajpurkar et al., 2017).

ii. Explainable AI (XAI).

   XAI techniques like Grad-CAM and saliency maps help visualize which regions of the X-ray influenced the model's prediction. This improves clinical trust and supports decision-making (Danilov et al., 2023; Tjoa & Guan, 2021).

iii. Chest X-ray Imaging.

   Chest radiographs are widely used in diagnosing pulmonary conditions. However, manual interpretation is prone to error, which motivates the use of AI-based systems.

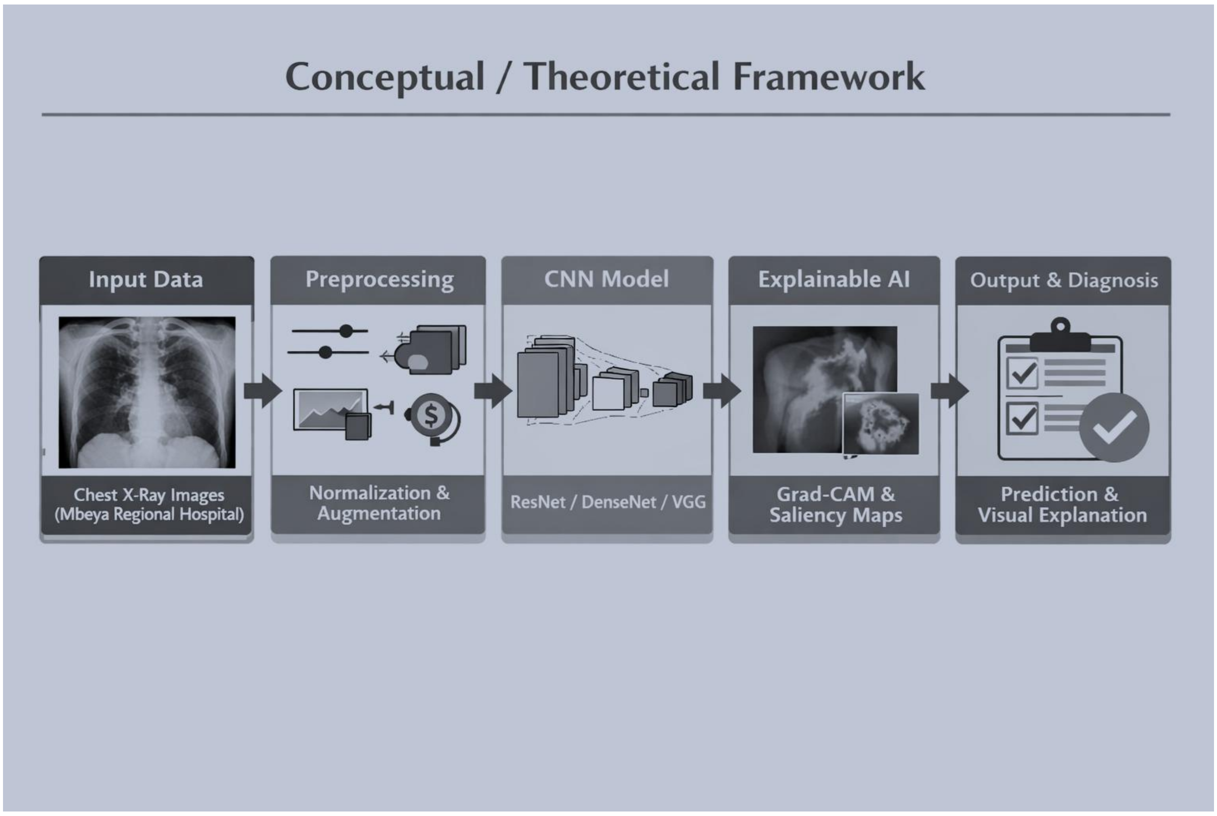

Figure 1: Conceptual/Theoretical Framework for Pulmonary Edema Detection

2.3 Review of Related Works

Several studies have explored the application of deep learning and explainable AI in medical imaging, particularly in the detection of pulmonary conditions. Rajpurkar et al., (2017) introduced CheXNet, a convolutional neural network trained on the ChestX-ray14 dataset, which achieved radiologist-level accuracy in pneumonia detection. Despite its success in accuracy, the model lacked interpretability, limiting its clinical adoption. Building on this, Irvin et al., (2019) developed the CheXpert dataset with uncertainty labels, enabling large-scale training for thoracic disease classification. Although this improved robustness, the uncertainty labels introduced complexity in model training. Danilov et al., (2024) advanced the field by integrating CNNs with Grad-CAM to visualize pulmonary edema features, thereby enhancing transparency and clinician confidence. However, their study emphasized interpretability more than accuracy optimization. Abbasi et al., (2025) focused on improving detection accuracy by employing ensemble CNNs with attention mechanisms, yet their approach did not incorporate explainability tools, leaving predictions as black-box outputs. Complementing these empirical works, Tjoa & Guan, (2021) provided a survey of explainable AI in medicine, underscoring the importance of transparency for clinical adoption, though their contribution was theoretical rather than dataset-driven.

Taken together, these studies demonstrate significant progress in applying CNNs to chest X-ray analysis. Rajpurkar and Abbasi prioritized accuracy, while Danilov emphasized interpretability, and Irvin contributed by providing large-scale datasets. Nevertheless, few studies have successfully balanced both accuracy and transparency, and most relied on international datasets rather than local hospital data. This creates a gap in contextual relevance for healthcare systems in regions such as Tanzania. While Danilov et al., (2024) addressed interpretability using Grad-CAM, their study did not fully integrate high-performance CNN architectures with explainability in a balanced manner, nor did it apply the approach to local datasets. This research therefore contributes by combining CNN models (ResNet, DenseNet, VGG) with Grad-CAM explanations, applied to chest X-rays from Mbeya Regional Hospital. In doing so, it ensures both diagnostic accuracy and clinical trust, while providing practical relevance to the Tanzanian healthcare context.

2.4 Empirical Review

2.4.1 Studies on Pulmonary Edema Detection

Abbasi et al., (2025) applied ensemble CNNs with attention mechanisms to detect pulmonary edema in chest X-rays. Their study achieved high accuracy but lacked explainability, limiting clinical trust. Danilov et al., (2023) trained CNNs on the MIMIC-CXR dataset to identify radiographic features such as Kerley lines and pleural effusions. Although effective in detection, their methodology did not integrate Explainable AI techniques, leaving predictions opaque.

2.4.2 Studies on Explainable AI in Medical Imaging.

Danilov et al., (2024) introduced Grad-CAM into their methodology, providing visual explanations of CNN predictions. This improved clinical adoption by allowing radiologists to verify AI decisions. Tjoa & Guan, (2021) conducted a survey on Explainable AI in healthcare, emphasizing the importance of transparency and interpretability in medical imaging systems.

2.4.3 Studies on Data and Model Robustness.

Esteva et al., (2019) highlighted the importance of large datasets in training robust AI models for healthcare applications. Shorten & Khoshgoftaar, (2019) demonstrated that data augmentation techniques improve generalization and reduce overfitting in deep learning models.

Table 2.1Empirical Review Methodology

Ref	Dataset Used	Year	Methodology	Findings	Limitations
Rajpurkar et al., (2017)	ChestX-ray14 (NIH, >100,000 images)	2017	CNN (CheXNet)	Achieved radiologist-level pneumonia detection.	No explainability, limited to pneumonia.
Abbasi et al., (2025)	Multi-hospital dataset	2025	Ensemble CNNs with attention.	Enhanced pulmonary edema detection accuracy.	No XAI integration, black-box predictions.
(Irvin et al., 2019)	CheXpert (>220,000 images)	2019	CNN with uncertainty labels.	Improved classification of multiple thoracic diseases	Uncertainty labels complicated training
Danilov et al., (2024)	Clinical chest X-rays (Oxford Academic study)	2024	Apply Grad-CAM	Visualized pulmonary edema features, improved clinician trust	Focused on interpretability, less on accuracy
Tjoa & Guan, (2021)	Survey of medical imaging studies	2020	Review of XAI methods	Highlighted importance of transparency in medical AI	Not dataset-based, theoretical survey

Strengths of Methodologies:

i. High accuracy in pulmonary edema detection.

ii. Use of large, diverse datasets.

iii. Application of augmentation to improve robustness.

Weaknesses of Methodologies:

i. Limited integration of explainability in most studies.

ii. Dependence on large datasets, which may not be available in resource-limited settings.

iii. Minimal clinical adoption due to lack of transparency.

2.5 Existing system

Several AI-based systems have been developed for chest X-ray analysis, focusing on the detection of pneumonia, tuberculosis, and other thoracic diseases. These systems demonstrate the potential of deep learning in medical imaging but reveal important limitations when applied specifically to pulmonary edema detection.

Examples of Existing Systems**:**

i. **CNN-based Systems;** Deep learning models such as ResNet, DenseNet, and VGG have been widely applied in chest X-ray analysis. Abbasi et al., (2025) used ensemble CNNs for pulmonary edema detection, achieving high accuracy but without explainability.

ii. CheXNet (Rajpurkar et al., 2017)**:** A CNN model trained on the ChestX-ray14 dataset to detect pneumonia. It achieved radiologist-level accuracy but lacked explainability.

iii. CheXpert (Irvin et al., 2019)**:** Expanded detection to multiple thoracic diseases, including pulmonary edema. While effective in classification, it still functioned as a "black-box" system.

iv. **Tuberculosis Detection System** (Lakhani & Sundaram, 2017)**:** CNN-based model with high accuracy, but no transparency in predictions.

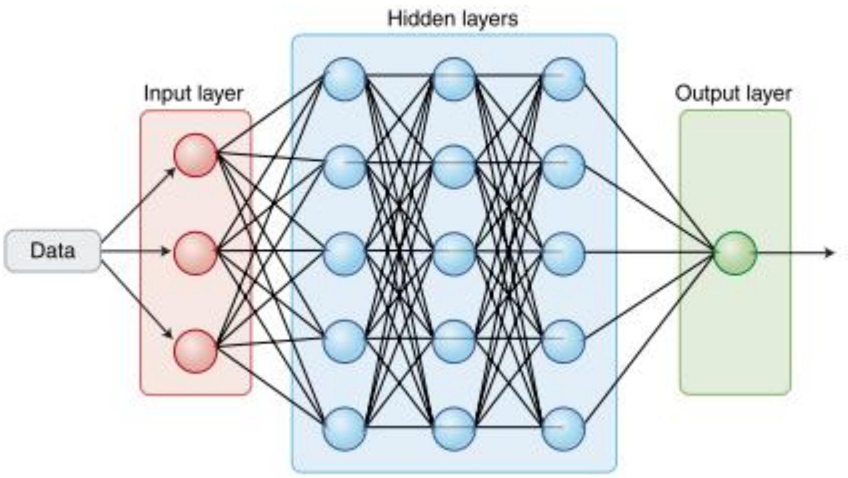

Figure 2.2 Existing system framework

The following are the drawbacks of the existing system ....

i. Limited focus on pulmonary edema compared to other diseases.

ii. Lack of integrated explainability features.

iii. Dependence on large datasets, which may not be available in resource-limited hospitals such as Mbeya Regional Hospital.

iv. Minimal clinical adoption due to lack of transparency.

2.6 Research Gap

The reviewed literature and existing systems demonstrate that deep learning, particularly Convolutional Neural Networks (CNNs), has achieved significant success in detecting thoracic diseases from chest X-ray images. Systems such as CheXNet and CheXpert have proven the potential of AI in medical imaging, while empirical studies confirm the effectiveness of CNNs in identifying radiographic features of pulmonary edema.

However, several gaps remain unaddressed:

Identified Gaps:

i. **Limited Focus on Pulmonary Edema** – Most existing systems emphasize pneumonia and tuberculosis, with pulmonary edema detection receiving minimal attention.

ii. **Lack of Explainability** – Current CNN-based systems operate as "black-boxes," providing predictions without transparency. Clinicians cannot easily verify how decisions are made, reducing trust and adoption.

iii. **Dependence on Global Datasets** – Many systems rely on massive datasets such as ChestX-ray14 or CheXpert, which do not reflect the diagnostic realities of local hospitals. In resource-limited settings like Mbeya Regional Hospital, such datasets are unavailable, making locally sourced data essential.

iv. **Minimal Clinical Adoption** – Despite high accuracy, the absence of integrated Explainable AI features has limited the use of these systems in real-world healthcare settings.

Key Findings:

i Deep learning systems are effective but not transparent.

ii Pulmonary edema detection is underexplored compared to other lung diseases.

iii There is a need for locally relevant solutions that can work with smaller datasets.

iv Integration of Explainable AI is essential to enhance clinical trust and usability.

Therefore, there is a need for a CNN-based system specifically designed for pulmonary edema detection, using chest X-ray data from Mbeya Regional Hospital, and integrated with Explainable AI techniques such as Grad-CAM. This approach will not only provide accurate diagnostic predictions but also highlight the regions of chest X-rays contributing to those predictions, thereby enhancing transparency, clinical trust, and usability in the Tanzanian healthcare context.

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